

RESEARCH INTERESTS

I am an infectious disease epidemiologist and mathematical modeller with over a decade of experience using data and models to support infectious disease control programmes in Indonesia and other low- and middle-income countries. I hold a PhD from Imperial College London, specialising in translating research into practical recommendations for public health policy. My work encompasses COVID-19 response, malaria and dengue control, and outbreak responses, utilising diverse data sources, including routine surveillance data, population surveys, and call detail records (mobile phone data). I have collaborated extensively with Indonesia's Ministry of Health and local government health departments, trained over 300 public health officers, secured competitive research funding including the MORU-OUCRU Discovery Research Academy Fellowship, and published research in leading medical and public health journals. Currently based at the National University of Singapore, I am committed to bridging academic research and real-world public health practice through innovative modelling approaches and sustained capacity building in Indonesia and across low- and middle-income countries.

EDUCATION

PhD in Epidemiology, Evolution, and Control of Infectious Diseases, Imperial College London 2019 - 2023

MRes in Biomedical Research (with distinction), Imperial College London 2018 - 2019

MSc in Epidemiology (with distinction), Imperial College London 2015 - 2016

BSc in Statistics, IPB University 2008 - 2012

EMPLOYMENT

Research Fellow, Saw Swee Hock School of Public Health, National University of Singapore 2024 - Present

Research Associate, MRC Centre for Global Infectious Disease Analysis, Imperial College London 2023 - 2024

Statistician, Eijkman-Oxford Clinical Research Unit (now: OUCRU Indonesia) 2014 - 2018

SELECTED PUBLICATIONS

Summary: 19 peer-reviewed publications | 6 first author | Citations: 1985 | h-index: 11

First Author

1. **Djaafara BA**, Adrian V, Eriawati E, Elyazar IRF, Hamers RL, Baird JK, Thwaites GE, Clapham HE (2026). Modeling the transmission dynamics and control strategies during the 2017 diphtheria outbreak in Jakarta, Indonesia. *Infectious Disease Modelling*. [link](#)
2. **Djaafara BA**, Elyazar IRF, Surya A, Silalahi FSM, Handito A, et al. (2025). Estimating dengue force of infection from age-stratified surveillance data in Java, Indonesia. *Journal of The Royal Society Interface*. [link](#)
3. **Djaafara BA**, Mutunga M, Eneanya OA, Forna A, Cucunubá ZM (2025). The bridge between two worlds: Global South researchers' journeys through Global North academic training and beyond. *Epidemics*. [link](#)
4. **Djaafara BA**, Sherrard-Smith E, Churcher TS, Fajariyani SB, Prameswari HD, Herdiana H, Puspawati RT, Lestari KD, Elyazar IRF, Walker PGT (2025). Spatiotemporal heterogeneity in malaria transmission across Indonesia: analysis of routine surveillance data 2010-2019. *BMC Medicine*. [link](#)
5. **Djaafara BA**, Elyazar IRF, Silalahi FSM, Surya A, Handito A, et al. (2025). Dengue transmission heterogeneity across Indonesia's archipelago: climate-driven spatiotemporal patterns and policy implications. *medRxiv*. **[Preprint]** [link](#)
6. **Djaafara BA**, Whittaker C, Watson OJ, Verity R, Brazeau NF, et al. (2021). Using syndromic measures of mortality to capture the dynamics of COVID-19 in Java, Indonesia, in the context of vaccination rollout. *BMC Medicine*. [link](#)
7. **Djaafara BA**, Imai N, Hamblin E, Impouma B, Donnelly CA, Cori A (2021). A Quantitative Framework for Defining the End of an Infectious Disease Outbreak: Application to Ebola Virus Disease. *American Journal of Epidemiology*. [link](#)

Co-Author

1. Surendra H, **Djaafara BA**, Prameswari HD, Supriyanto D, Waluyo P, et al. (2024). Mitigating risks of malaria and other vector-borne diseases in the new capital city of Indonesia. *Nature Communications*. [link](#)
2. Johnson R, **Djaafara B**, Haw D, Doohan P, Forchini G, Pianella M, Ferguson N, Smith PC, Hauck KD (2023). The societal value of SARS-CoV-2 booster vaccination in Indonesia. *Vaccine*. [link](#)
3. Fadilah I, **Djaafara BA**, Lestari KD, Fajariyani SB, Sunandar E, et al. (2022). Quantifying spatial heterogeneity of malaria in the endemic Papua region of Indonesia: Analysis of epidemiological surveillance data. *The Lancet Regional Health - Southeast Asia*. [link](#)
4. Surendra H, Salama N, Lestari KD, Adrian V, Widyastuti, Oktavia D, Lina RN, **Djaafara BA**, et al. (2022). Pandemic inequity in a megacity: a multilevel analysis of individual, community and healthcare vulnerability risks for COVID-19 mortality in Jakarta, Indonesia. *BMJ Global Health*. [link](#)
5. Whittaker C, Walker PGT, Alhaffar M, Hamlet A, **Djaafara BA**, Ghani AC, Ferguson N, Dahab M, Checchi F, Watson OJ (2021). Under-reporting of deaths limits our understanding of true burden of covid-19. *BMJ*. [link](#)
6. Surendra H, Elyazar IRF, **Djaafara BA**, Ekawati LL, Saraswati K, et al. (2021). Clinical characteristics and mortality associated with COVID-19 in Jakarta, Indonesia: A hospital-based retrospective cohort study. *The Lancet Regional Health - Western Pacific*. [link](#)

SELECTED PRESENTATIONS

- Modelling to inform policymaking. *Collaborative workshop on Infectious Disease Modelling for Health Policy and Research*, Oct 2025
- Subnational dengue risk mapping of Indonesia. *Dengue multisource collaborative surveillance workshop (WHO Indonesia & Indonesia Ministry of Health)*, Apr 2025
- Modelling to inform policymaking: experience from Indonesia. *Modelling and inference for pandemic preparedness workshop, Isaac Newton Institute*, Aug 2024

- Modelling for dengue: estimating burden, intervention impact, and scenario simulations. *Dengue multisource collaborative surveillance workshop (WHO SEARO & Indonesia Ministry of Health)*, Jul 2024
- Using models and routine data to measure progress towards malaria elimination in Papua, Indonesia. *MIDSEA online seminar series*, Aug 2023
- My personal experience in modelling during public health emergency'. *MIDSEA workshop: Modelling during a public health emergency*, May 2023
- Subnational estimates of dengue force of infection and burden in Indonesia. *Infectious Disease Modelling (IDM) Conference*, Nov 2024 [**Poster**]
- A decade of malaria control in Indonesia (2010-2019): analysis of routine surveillance and bed nets distribution data. *American Society of Tropical Medicine and Hygiene (ASTMH) Conference*, Nov 2022 [**Poster**]
- A quantitative framework to define the end of an outbreak: application to Ebola Virus Disease. *EPIDEMICS7*, Dec 2019 [**Poster**]

GRANTS & AWARDS

- **MORU-OUCRU Discovery Research Academy (MODRA)**, Wellcome (2025 - 2026)
- **Travel grant**, WHO Hub for Pandemic and Epidemic Intelligence (2024)
- **Early career researcher bursary**, Isaac Newton Institute, University of Cambridge (2024)
- **Student Awards for Outstanding Achievement Medal**, Imperial College London (2022)
- **MRes + PhD Studentship**, School of Public Health, Imperial College London (2018)

TEACHING

- Mathematical modelling of infectious diseases: (why + how) we do it? (Guest Lecturer), Monash University Indonesia, University of Indonesia
- Introduction to Infectious Disease Modelling (Teaching Assistant), Imperial College London
- Introduction to Statistical Thinking and Data Analysis (Teaching Assistant), Imperial College London
- Modelling and Analysis of Serological Data: An Introductory Short Course, Imperial College London & OUCRU Vietnam
- Basics of Model Fitting, MIDSEA Summer School - Inference Track
- Basics of Model Fitting, MIDSEA Summer School - Inference Track
- Infectious Disease Transmission Modelling for Mitigating Potential Epidemic Threats, 2nd Gadjah Mada International Conference of Tropical Medicine (GAMA - ICTM)

SKILLS

Programming: R, Stan | **Methods:** Bayesian inference, Infectious disease modelling, Statistical modelling | **Languages:** English, Indonesian

REFERENCES

Dr Swapnil Mishra (Saw Swee Hock School of Public Health, National University of Singapore) | Dr Patrick Walker (Department of Infectious Disease Epidemiology, Imperial College London) | Dr Iqbal Elyazar (Oxford University Clinical Research Unit (OUCRU) Indonesia)
Contact details available upon request